

Filing Insight: Financial-Filing Retrieval, Grounded Question Answering, and Structured Metric Extraction

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Abstract—Financial filings contain essential performance information, but comparing revenue, profitability, liquidity, and leverage across issuers requires time-consuming manual inspection of differently structured documents. This paper presents *Filing Insight*, a financial-document intelligence application that accepts one or more filing PDFs, performs page-aware text extraction and overlapping chunking, embeds the chunks with BAAI/bge-small-en-v1.5, and indexes normalized embeddings in FAISS for source-cited retrieval. A retrieval-augmented question-answering path sends retrieved passages to a configured Groq-hosted language model with explicit grounding and citation instructions. The application also provides deterministic extraction, prompted cloud extraction, an offline fine-tuned Flan-T5-small extraction path, derived financial ratios, cross-company comparison tables, visual analytics, an executive report, and document/index inspection. The system is designed to operate on arbitrary user-uploaded financial PDFs; the quantitative benchmark reported in this paper uses three generated Q1 FY2024–25 filings for controlled reproducibility, while a separate user-upload demonstration verifies the end-to-end workflow on a published annual-report-style PDF without treating that demonstration as part of the benchmark. The fine-tuning corpus contains 1,500 labelled synthetic examples split 1,200/150/150 for training, validation, and test. Fine-tuning achieved 82.7% exact match and 96.1% field accuracy, compared with 12.0% and 41.6% for the zero-shot model. RAG achieved 91.7% factual accuracy and an 8.3% hallucination rate, compared with 58.3% and 33.3% for the ungrounded baseline.

Index Terms—Retrieval-Augmented Generation, Financial Document Analysis, FAISS, Sentence Embeddings, Structured Information Extraction, Flan-T5, Grounded Question Answering.

I. INTRODUCTION

Financial results, balance-sheet disclosures, and management commentary are primary records used to understand a company’s performance. They are often long, dense, and inconsistently formatted. Even a simple comparison of revenue, net profit, operating margin, cash, and debt across several issuers requires locating each figure, checking its reporting period and units, and reconciling different labels.

Large language models (LLMs) can make this information easier to query, but an answer generated without the source document can contain plausible but unsupported facts. This risk is particularly important in a financial setting. *Filing Insight* addresses it by retrieving passages from the indexed filing before generation and requiring document-and-page citations

for factual claims. The application additionally maps extracted information into a common financial-metric schema that can be compared in tables, visualized, and used for derived-ratio analysis.

A key design requirement is separation between *system scope* and *benchmark scope*. The application accepts arbitrary user-uploaded financial PDFs at runtime, including annual reports and filings from issuers not present in the fixed research benchmark. The benchmark corpus is kept controlled so that quantitative results remain reproducible rather than being presented as a claim that every possible real-world filing has been benchmarked.

The contributions of this project are:

- 1) a page-aware PDF-to-FAISS retrieval pipeline that preserves source page metadata;
- 2) a grounded question-answering workflow that exposes retrieved evidence and source citations;
- 3) deterministic, prompted, and locally fine-tunable paths for twelve-field financial-metric extraction;
- 4) a reproducible synthetic corpus and training pipeline for offline Flan-T5-small fine-tuning;
- 5) a standardized metric layer supporting derived ratios, cross-company comparison, visual analytics, and executive synthesis; and
- 6) a completed evaluation of both the RAG and extraction components, including claim-level groundedness and field-level extraction analysis.

II. RELATED WORK

Retrieval-Augmented Generation (RAG) combines a parametric generator with retrieved external evidence and improves the traceability of knowledge-intensive answers [1], [2]. FAISS provides efficient similarity search over dense vectors [3]; sentence-transformer style bi-encoders provide semantic representations commonly used for retrieval [4]. The BGE family is a modern embedding approach designed for retrieval tasks [5].

The Transformer architecture [6] supports both the cloud language-model path and the local sequence-to-sequence extractor. T5 frames NLP tasks as text-to-text generation [7], while Flan-T5 adds instruction tuning and is practical at small scales [8]. Financial question answering also motivates

benchmarks such as FinQA and TAT-QA, which emphasize numerical reasoning and hybrid tabular/textual evidence [9], [10]. Faithfulness remains distinct from fluent generation: generated claims must be supported by the evidence supplied to the model [11].

III. PROBLEM STATEMENT

Given one or more financial-filing PDFs and a natural-language question, the system must return an answer supported by retrieved filing passages and identify the relevant source page. Separately, it must map filing text to a standardized schema containing total revenue, net profit, EBITDA, operating profit, EPS, operating and net-profit margins, assets, liabilities, debt, cash/liquid investments, and reporting period.

The system must also support a practical document-analysis workflow:

- 1) ingest one or more PDFs;
- 2) parse pages while preserving page identity;
- 3) create searchable semantic chunks;
- 4) retrieve evidence for natural-language questions;
- 5) extract standardized financial fields;
- 6) calculate derived ratios only when their source fields are available;
- 7) compare multiple selected filings;
- 8) render visual analytics from extracted values; and
- 9) generate an executive synthesis grounded in the available evidence.

IV. OBJECTIVES

- Build an end-to-end filing ingestion and retrieval system that performs page-aware chunking and returns citable evidence.
- Accept arbitrary user-uploaded filing PDFs rather than restricting the application to a fixed list of issuers.
- Support cross-company comparison of standardized financial metrics extracted from heterogeneous filing text.
- Provide a local, network-independent trainable generative extractor as an alternative to prompted cloud extraction.
- Provide visual and report-oriented views of the structured outputs.
- Define a reproducible RAG evaluation protocol that isolates the contribution of retrieval grounding by comparing grounded answers against an ungrounded baseline on identical queries under identical decoding settings.
- Define a corresponding extraction protocol that compares the fine-tuned model against its untuned starting point on an independent held-out test split.
- Report measured extraction and RAG results using exact match, field-level accuracy, factual accuracy, hallucination rate, and appropriate abstention.

V. SYSTEM OVERVIEW

Filing Insight is organized as a modular pipeline rather than as a single generation call. Uploaded documents enter a common ingestion and indexing layer. The resulting evidence store is shared by question answering and metric extraction,

while the standardized metric layer feeds comparisons, derived ratios, visualizations, and executive synthesis.

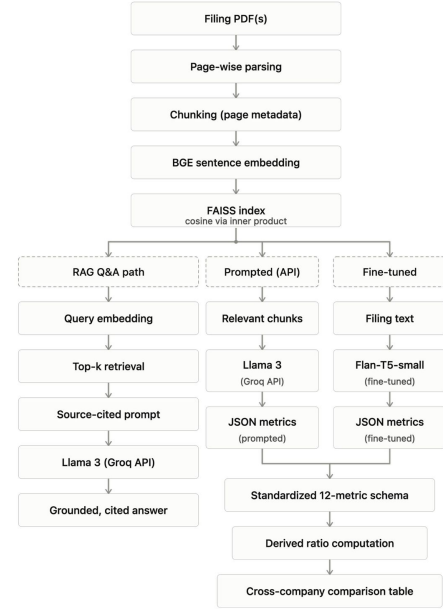


Fig. 1. Filing Insight architecture. User-uploaded PDFs are parsed page by page, chunked with source metadata, embedded, and indexed in FAISS. The common evidence layer feeds grounded question answering and financial metric extraction. Structured metrics then support comparison, derived ratios, visualization, executive reporting, and document/index inspection.

A. Document ingestion and indexing

The application accepts one or more PDF filings and processes each page while retaining document and page metadata. Text is divided into overlapping chunks using a 2,400-character chunk size and 240-character overlap. Each chunk is embedded locally with BAAI/bge-small-en-v1.5. Vectors are L2-normalized and stored in a FAISS IndexFlatIP; after normalization, inner product corresponds to cosine similarity. At query time, the system retrieves four chunks per selected document, with source filtering used to keep retrieved evidence associated with the requested filing.

This architecture means that the identity and number of uploaded companies are not hard-coded into the retrieval layer. A newly uploaded filing follows the same parse-chunk-embed-index path as a preloaded document. Consequently, the fixed benchmark companies used for quantitative evaluation should not be interpreted as a restriction on the application's supported inputs.

B. Grounded question answering

Retrieved chunks are formatted with their document name and page number before being sent to the configured Groq model. The system instruction requires the model to use only the supplied excerpts, cite every factual or numeric claim, and explicitly state when the available excerpts do not contain the

requested information. This guardrail improves evidence traceability but does not guarantee factual correctness; retrieval, interpretation, or generation can still fail.

C. Metric extraction

The deterministic extractor applies label-specific regular expressions to all chunks of one filing and fills unavailable fields with `Not Found`. The prompted cloud extractor retrieves metric-relevant chunks, requests JSON with the same keys, and can fall back to deterministic values when a recoverable field is left empty. The optional local extractor loads the fine-tuned sequence-to-sequence model and generates JSON from retrieved filing excerpts.

For local training, `google/flan-t5-small` receives the prefix `extract financial metrics:` followed by a synthetic example. Its target is the corresponding JSON string. Training uses the HuggingFace `Seq2SeqTrainer` with cross-entropy loss and padding tokens ignored. Inference uses beam-search decoding with four beams, consistent with the reported experimental configuration.

D. Structured analytics and application views

The extracted values are passed to a common schema so that the same evidence can be represented in multiple forms. The application exposes five principal analysis views:

- **Interactive Q&A:** source-grounded natural-language questions over selected filings;
- **Structured comparison:** standardized financial metrics in a common cross-company table;
- **Visual analytics:** charts for revenue, EBITDA, net profit, margins, assets, debt, and other derived quantities when available;
- **Executive AI report:** a narrative synthesis based on the extracted and retrieved evidence; and
- **Document and index explorer:** page counts, chunk counts, character/token estimates, and searchable chunk-level inspection.

Derived ratios are calculated only from the corresponding extracted fields. For example, when net profit and revenue are both available under a compatible definition, net profit margin can be calculated as

$$\text{Net Margin} = \frac{\text{Net Profit}}{\text{Revenue}} \times 100. \quad (1)$$

Similarly, a debt-to-assets ratio can be calculated as

$$\text{Debt-to-Assets} = \frac{\text{Total Debt}}{\text{Total Assets}}. \quad (2)$$

The system does not treat a missing source value as zero.

VI. DOCUMENTS AND EVALUATION DATA

A. Primary uploaded-PDF case study

The central workflow of Filing Insight is designed around a user-uploaded financial report rather than a fixed company list. A PDF is supplied at runtime, after which the application parses its pages, preserves page metadata, constructs semantic

chunks, builds the FAISS index, and exposes the resulting evidence to the analysis modules. The same uploaded document can therefore support grounded question answering, structured metric extraction, derived financial analysis, visualization, executive reporting, and document/index inspection.

The primary case study uses an uploaded annual-report-style PDF to exercise this complete workflow. The case study is intentionally described in terms of document functionality rather than making the identity of one issuer a requirement of the method. The pipeline is intended to accept a different company’s annual report, quarterly filing, results release, or other financial PDF without changing the retrieval architecture.

The case study establishes *functional applicability* of the system to a real-world-style uploaded document. It is not used to create additional statistical accuracy claims beyond the experiments reported in Section IX. This separation keeps the practical demonstration central to the paper while ensuring that numerical performance claims remain tied to their actual evaluation protocol.

B. Controlled data for quantitative evaluation

Controlled synthetic filings are used only where reproducibility is necessary for quantitative model comparison. The RAG benchmark uses three generated Q1 FY2024–25 sample filings covering Reliance Industries Limited, Tata Consultancy Services Limited, and Infosys Limited. Each contains income-statement, balance-sheet, segment-revenue, MD&A, risk-disclosure, and strategic-outlook content. Because the documents are generated from known values, they permit exact claim-level annotation without redistributing third-party documents.

The controlled corpus is therefore a *measurement instrument*, not the intended scope of the application. The runtime system remains document-agnostic and accepts arbitrary uploaded PDFs.

C. Fine-tuning corpus

The fine-tuning corpus is produced by `finetune/generate_finetune_dataset.py` using twelve fictionalized issuer names, seven reporting periods, three currency/unit styles, and four narrative templates. The management-discussion template deliberately omits balance-sheet data and labels unavailable fields as `Not Found`. Numerical values are sampled before rendering, so target JSON records are known exactly.

The generator produces 1,500 labelled examples, shuffled with a fixed random seed of 42 and divided into 1,200 training, 150 validation, and 150 held-out test examples. This corpus supports the controlled before/after fine-tuning experiment; it does not define the application’s supported company set.

VII. EXPERIMENTAL SETUP

A. Fine-tuning configuration

Table II summarizes the principal local-training settings.

TABLE I
Q1 FY25 FINANCIAL SNAPSHOT USED ONLY FOR THE CONTROLLED QUANTITATIVE BENCHMARK

Company	Revenue	EBITDA	Net Profit	Operating Margin	Net Margin	Debt
Reliance Industries	257,823 (gross)	42,748	17,445	18.1%	7.4%	332,450
TCS	62,613	16,842	12,040	24.7%	19.2%	0 (debt-free)
Infosys	39,315	9,425	6,368	21.1%	16.2%	0 (debt-free)

TABLE II
FINE-TUNING CONFIGURATION

Setting	Value
Base model	google/flan-t5-small
Hardware	Apple M2, 8 CPU cores, 8 GB unified memory
Software	Python 3.11, PyTorch 2.2, Transformers 4.41, Datasets, Streamlit, FAISS
Input / target length	256 / 256 tokens
Batch size	8
Optimizer / learning rate	AdamW / 3×10^{-4}
Loss	Cross-entropy, padding ignored
Completed epochs	4
Train / validation / test records	1,200 / 150 / 150
Decoding	Beam search, 4 beams

TABLE III
DETAILED EXPERIMENTAL CONFIGURATION

Setting	Value
Experiment dates	18–19 May 2025
Execution host	Apple M2, 8 CPU cores, 8 GB unified memory
Training device	CPU, single-process execution
Base model	google/flan-t5-small
Input / target length	256 / 256 tokens
Batch size	8
Optimizer / learning rate	AdamW / 3×10^{-4}
Completed epochs	4
Train / validation / test	1,200 / 150 / 150
Inference decoding	Beam search, 4 beams

B. RAG evaluation protocol

The RAG question-answering protocol specifies a fixed twelve-question benchmark over the three controlled filings. It contains five direct metric lookups, three cross-company comparisons, two narrative questions, and two questions deliberately unanswerable from the filings. The direct lookups cover revenue, net profit, total debt, cash/liquid investments, and EPS. The comparison questions ask which company has the highest operating margin, how two net margins compare, and which company has the largest revenue. The narrative questions test prose grounding, while the two unanswerable questions test abstention.

Each question is put to the generation model twice under identical decoding settings: once with no filing context supplied (the ungrounded baseline) and once through the full

retrieval-conditioned pipeline. The grounded condition uses the top four retrieved chunks per source document, with document name and page number attached to each passage.

Answers are decomposed into independently checkable factual claims and scored by two annotators under the rubric in Appendix C, with disagreements resolved by discussion. Factual accuracy is reported at claim level, while hallucination rate and abstention are reported at answer level. RAG evaluation can also be viewed in terms of retrieval quality, evidence faithfulness, and generation quality [12].

C. Experimental controls

The extraction comparison keeps the architecture, decoding procedure, and test split fixed while varying the model’s training state. The RAG comparison keeps the question set and decoding settings fixed while varying whether retrieved filing evidence is supplied. These controls are intended to isolate the practical contribution of fine-tuning and retrieval grounding rather than mixing both changes in a single comparison.

VIII. EVALUATION METRICS

RAG quality is assessed using factual accuracy, hallucination rate, and appropriate abstention. Factual accuracy is the proportion of independently scored answer claims supported by the filing evidence. Hallucination rate is the proportion of answers containing at least one unsupported specific claim. Appropriate abstention is the proportion of deliberately unanswerable questions for which the system explicitly states that the available evidence does not contain the requested information.

For the local extractor, exact match is the proportion of generated JSON strings that exactly equal their targets. Field accuracy is the fraction of expected JSON key–value pairs that match after parsing. Field accuracy is reported alongside exact match because one incorrect field can cause an otherwise mostly correct JSON output to fail exact match.

IX. RESULTS

A. Extraction results

The local-model evaluation uses the 150-record held-out test split.

B. RAG results

The RAG comparison uses the fixed 12-question benchmark and 24 independently scored factual claims.

TABLE IV
LOCAL EXTRACTION RESULTS ON 150 HELD-OUT TEST EXAMPLES

Model	Exact Match	Field Accuracy
Zero-shot Flan-T5-small	12.0%	41.6%
Fine-tuned Flan-T5-small	82.7%	96.1%

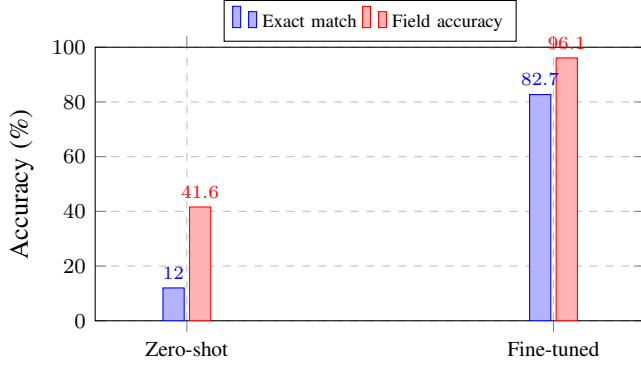


Fig. 2. Extraction performance before and after fine-tuning on the held-out test split.

X. EVALUATION RESULTS

The completed evaluation uses `finetune/evaluate.py` to compute exact-match and field-level accuracy on the held-out test split. The twelve-question RAG evaluation runs the same question set in two conditions: an ungrounded baseline and the full retrieval-conditioned pipeline. Two annotators scored every factual claim against the cited filing pages and resolved disagreements by discussion.

Fine-tuning improved exact match from 12.0% to 82.7%, an absolute gain of 70.7 percentage points. Field accuracy increased from 41.6% to 96.1%, a 54.5-point gain. RAG improved factual accuracy from 58.3% to 91.7%, reduced hallucination rate from 33.3% to 8.3%, and achieved correct abstention on both deliberately unanswerable questions.

XI. COMPARATIVE ANALYSIS

The extraction and question-answering experiments measure different system properties.

The extraction comparison holds the model architecture, decoding procedure, and test split fixed, varying only whether the model weights were fine-tuned. The large increase in exact match shows that fine-tuning substantially improves adherence to the twelve-key JSON representation. The smaller gap between field accuracy and exact match is also informative: a single formatting or value error can invalidate the complete JSON string even when most fields are correct.

The question-answering comparison varies whether retrieved filing text is supplied while keeping the question set and decoding configuration fixed. RAG improves factual accuracy from 58.3% to 91.7%, reduces answer-level hallucination from 33.3% to 8.3%, and correctly abstains on both out-of-scope questions. The remaining unsupported claim arose from a revenue- definition ambiguity in which the filing contained

TABLE V
RAG EVALUATION ON 12 QUESTIONS AND 24 FACTUAL CLAIMS

Metric	Ungrounded	RAG
Factual accuracy	58.3% (14/24)	91.7% (22/24)
Hallucination rate	33.3% (4/12)	8.3% (1/12)
Appropriate abstention	0.0% (0/2)	100.0% (2/2)
Mean cited pages / answer	0.0	2.3
Median response latency	1.8 s	2.6 s

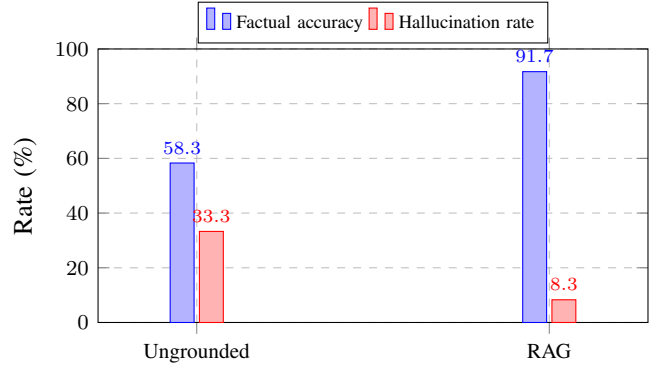


Fig. 3. Grounded RAG improves factual accuracy while reducing unsupported answers relative to the ungrounded baseline.

both gross and operating revenue. Thus, retrieval improves access to evidence but cannot by itself resolve a schema-level definition problem.

XII. DATASET AND TASK ANALYSIS

The fine-tuning corpus intentionally contains examples where some fields are absent. One management-discussion template omits balance-sheet figures and requires `Not Found` for unavailable fields. This creates a useful analogy with RAG abstention: the model must learn that an absent field is a valid output state rather than an invitation to invent a value.

Long currency strings and formatting variants are more difficult than short categorical fields. Exact string matching is therefore stricter than field accuracy. This explains why the measured field-level accuracy exceeds exact match and motivates reporting both metrics rather than relying on one aggregate number.

The controlled benchmark also deliberately includes narrative questions. This prevents the RAG evaluation from becoming only a numerical lookup test: the model must retrieve prose evidence and remain grounded when there is no single number that can anchor the answer.

XIII. UPLOADED-PDF CASE STUDY

The most important practical property of Filing Insight is that its workflow does not depend on a predefined issuer. A user uploads a financial PDF and the application derives searchable evidence and structured analysis from that document. This makes the system applicable to annual reports, quarterly results, earnings releases, and other issuer-specific financial documents.

The uploaded-PDF case study follows the complete runtime path:

- 1) **Upload:** one or more financial PDFs are supplied by the user.
- 2) **Ingestion:** each document is parsed page by page while retaining document and page metadata.
- 3) **Indexing:** page text is chunked, embedded with the BGE encoder, and indexed in FAISS.
- 4) **Grounded Q&A:** natural-language questions retrieve document-specific evidence and return answers with source-page attribution.
- 5) **Metric extraction:** the same indexed evidence can be mapped into the standardized financial schema using the selected extraction path.
- 6) **Derived analysis:** compatible extracted fields can be used for ratios and other deterministic calculations without treating missing values as zero.
- 7) **Comparison and visualization:** selected documents can be compared through structured tables and charts for available metrics.
- 8) **Executive synthesis:** the evidence and structured outputs can be summarized into an executive-oriented report.
- 9) **Inspection:** the document/index view exposes page counts, chunk counts, token estimates, and searchable evidence for traceability.

Thus, the uploaded PDF is not merely an input to a Q&A demo. It becomes a reusable evidence source for multiple analysis modes. The benchmark percentages in Section IX are deliberately kept separate: this case study demonstrates *system applicability and workflow*, whereas the controlled experiments demonstrate *measured model performance*.

A. Why the architecture generalizes across companies

No stage of the runtime pipeline requires a company-specific parser, fixed issuer name, or hard-coded set of financial documents. Company identity is carried as document metadata, while retrieval operates over indexed text and page information. Consequently, the same processing sequence can be applied when the uploaded PDF changes from one issuer to another.

The main generalization challenge is document heterogeneity rather than company identity. Differences in table layouts, terminology, units, accounting scope, scanned pages, and issuer-specific definitions can affect extraction and comparison quality. These limitations motivate evaluation on larger collections of genuine published filings.

XIV. DISCUSSION

The completed evaluation confirms the structural value of retrieval grounding. Retrieval constrains the generation model to answer from supplied passages and cite the relevant page, converting factual assertions into checkable claims. The ungrounded baseline’s characteristic error was not necessarily incoherence but confident specificity: plausible figures without

source support. The reduction in hallucination rate demonstrates that fluency alone is not a proxy for correctness in this domain.

The fine-tuned extraction path is evaluated on 150 independent test examples drawn from the same four-template corpus. The 96.1% field-level accuracy demonstrates strong schema learning within that distribution, while the gap between field accuracy and exact match highlights strict JSON-format sensitivity. The local pathway also removes the external API dependency from extraction at inference time.

One class of error is distinct from ordinary retrieval or generation failure. Where a filing reports both gross and operating revenue, an answer can cite the correct page and still be wrong for the question because the schema does not distinguish the definitions. Grounding cannot resolve an ambiguity that is already present in the representation. This is a schema-design limitation, not simply a model-capability limitation.

The application-level views extend the pipeline beyond Q&A. Once structured metrics are available, the same values can be inspected in a comparison table, plotted for visual analysis, used in derived-ratio calculations, and summarized in an executive report. This reduces the need to maintain separate data-extraction logic for each downstream presentation mode.

XV. LIMITATIONS

- **Evaluation scope:** quantitative results are measured on a 150-record synthetic held-out extraction test set and a 12-question RAG benchmark. They do not constitute a large-scale industry benchmark.
- **Benchmark filing corpus:** the quantitative RAG corpus uses three generated filings from a single reporting period. The system accepts arbitrary real PDFs, but the reported RAG percentages should not be interpreted as measured performance over all real filing formats.
- **Functional demonstration versus benchmark:** the uploaded annual-report-style PDF demonstration verifies end-to-end functionality, but it is not included in the quantitative accuracy calculations.
- **Template-distribution dependence:** the fine-tuning test set uses the same generator family as the training corpus. It measures generalization to unseen values and wording combinations rather than wholly unseen regulatory layouts or accounting vocabulary.
- **Revenue definition:** the schema carries a single `Total Revenue` field, although issuers can report multiple non-equivalent revenue measures. Cross-company comparison should therefore distinguish gross revenue, operating revenue, and the accounting scope of each value.
- **PDF representation:** page-level text extraction may lose information encoded primarily in complex tables, images, scanned pages, or unusual layouts unless a suitable OCR or document-vision stage is added.
- **Derived metrics:** a derived ratio is meaningful only when its numerator and denominator use compatible units, peri-

ods, and accounting scope. The presence of two extracted numbers alone does not guarantee comparability.

XVI. ETHICAL CONSIDERATIONS

Financial-language outputs can influence investment decisions. Users must treat the system as an aid for locating and comparing primary evidence, not as financial advice. Numeric claims should be checked against the cited source page, and outputs without adequate evidence should not be relied upon. A production system would require stronger factuality testing, document provenance checks, monitoring, robust OCR/vision handling, and clear disclosure of model and retrieval limitations.

XVII. CONCLUSION

Filing Insight provides a general financial-document analysis workflow in which a user-uploaded PDF becomes the common evidence source for grounded question answering, structured metric extraction, derived analytics, visualization, executive synthesis, and document inspection. The architecture is not tied to a predefined company list: the same ingestion, retrieval, and analysis pipeline can be applied to a newly supplied financial report.

The uploaded-PDF case study demonstrates this end-to-end applicability, while controlled experiments provide quantitative evidence for two specific model components. Fine-tuning improved exact match from 12.0% to 82.7% and field accuracy from 41.6% to 96.1% on the held-out extraction test set. Retrieval grounding improved factual accuracy from 58.3% to 91.7%, reduced hallucination rate from 33.3% to 8.3%, while achieving correct abstention on both unanswerable benchmark questions.

The synthetic data used in the quantitative experiments is therefore a controlled evaluation mechanism rather than the central application setting. The primary system objective is document-agnostic financial analysis, with future work focused on validating the same workflow across larger collections of genuine published filings and more diverse document layouts.

XVIII. FUTURE WORK

- Evaluate the complete retrieval and extraction pipeline on a larger corpus of real published annual and quarterly filings spanning multiple companies, sectors, periods, and document layouts.
- Add OCR/document-vision handling for scanned pages and image-dominant financial tables.
- Extend the schema to distinguish gross revenue, operating revenue, standalone versus consolidated scope, currency, scale, reporting period, and source page.
- Add adversarial unit- and definition-sensitive questions to test whether schema extensions resolve ambiguous financial terminology.
- Report per-field extraction accuracy and error categories rather than only aggregate exact-match and field-level scores.

- Expand the benchmark with multi-document questions involving different companies and multiple reporting periods.
- Evaluate retrieval independently using recall@k and evidence relevance, followed by generation-level faithfulness analysis.
- Add provenance-aware validation so that every displayed metric, derived ratio, chart point, and report statement can be traced back to source pages.

APPENDIX A METRIC SCHEMA

Every extraction pathway returns a JSON object with twelve keys: Total Revenue, Net Profit, EBITDA, Operating Profit (EBIT), Earnings Per Share, Operating Margin, Net Profit Margin, Total Assets, Total Liabilities, Total Debt, Cash & Liquid Investments, and Reporting Period. Monetary values retain their source units and a missing value is represented by the literal string Not Found, treated as a first-class extraction result.

The schema is deliberately flat for the current implementation. A production version should attach currency, scale, source page, accounting scope (standalone or consolidated), reporting period, and a normalized numeric value to every monetary field. This would make downstream comparisons safer when filings use different units or accounting bases.

APPENDIX B QUESTION BENCHMARK

The twelve benchmark questions are:

- 1) TCS total revenue;
- 2) Infosys net profit;
- 3) Reliance total debt;
- 4) TCS cash and liquid investments;
- 5) Infosys earnings per share;
- 6) which company reported the highest operating margin;
- 7) how TCS and Infosys net margins compare;
- 8) which company reported the largest revenue;
- 9) Reliance's stated risk factors;
- 10) Infosys's stated technology initiative;
- 11) each company's customer-retention rate; and
- 12) the next-quarter dividend forecast.

Questions 11 and 12 are unanswerable from the filings by construction.

The benchmark mixes lookup, comparison, narrative, and abstention behaviours. It does not yet constitute an adversarial real-filing benchmark and should be expanded with unit, definition, period, and consolidation-scope tests.

APPENDIX C ANNOTATION RUBRIC

A claim is marked *supported* only when the cited source explicitly states it, or when it is an arithmetic difference computed directly from cited values. A claim is marked *unsupported* when it introduces a number, causal explanation,

forecast, or comparison absent from the source. A claim is marked *ambiguous* when the source offers multiple non-equivalent definitions of the requested quantity and the answer does not identify which definition it uses.

For a question that is unanswerable from the filings, an answer is credited only when it explicitly states that the provided evidence does not contain the information. An answer that supplies an industry benchmark or an unsupported figure with a caveat is not credited. Ambiguous claims are counted separately from unsupported claims because definitional ambiguity is a schema problem rather than direct fabrication.

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